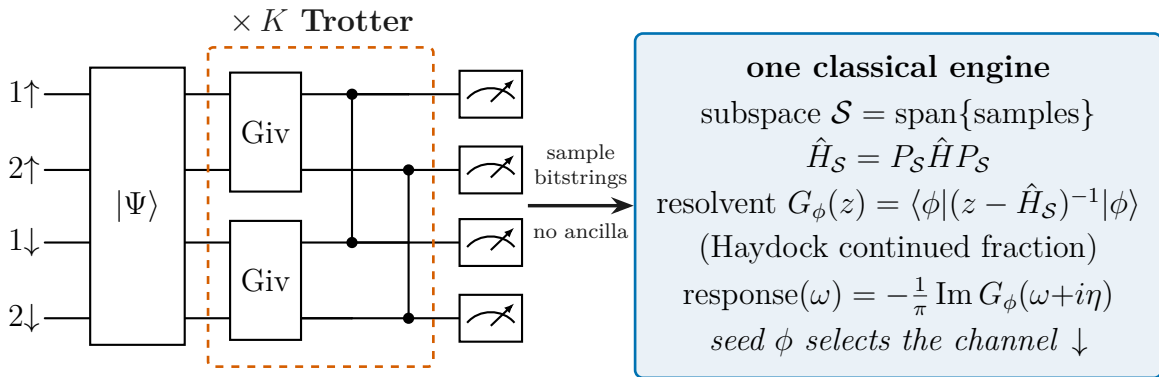




$\vdots$ 12 qubits ($L=6 \times \text{spin}$); $\text{Giv} \equiv \text{hop } t$, $R_{zz} \equiv U \hat{n}_{\uparrow} \hat{n}_{\downarrow}$

single particle ($N \pm 1$)

seed $\phi = \hat{c}_p^{\dagger} |0\rangle$, $\hat{c}_p |0\rangle$

$\Rightarrow A(\omega)$, $A(k, \omega)$

photoemission / ARPES

collective (N , neutral)

seed $\phi = \hat{n}_q |0\rangle$, $\hat{S}_q^z |0\rangle$

$\Rightarrow S(q, \omega)$, $S^{zz}(q, \omega)$

neutron / X-ray scattering